Anh Dao

anhdd2000@gmail.com | +358 41 325 1677 | daoa0601.github.io

EDUCATION

M.Sc. (Tech.) in Computer Science, Aalto University — ***Aug 2024 – Jul 2026***
Graduated 31 July 2026. GPA: 4.09/5.00. Minor: Machine Learning, Data Science and AI
Thesis: *Mitigating Sensor-Constrained Attacks on Virtual Reality Cybersickness Classifiers*

B.A. in Computer Science, Grinnell College (GPA 3.76/4.00) — ***Aug 2019 – May 2023***

EXPERIENCE

Smartly – ***Software Engineer*** — ***Feb 2026 – Present***

- Working on the Agentic Orchestration team for Smartly's core initiative Synapse, across **Python**/**Pydantic AI** and **TypeScript**/**Node.js** projects.
- Building Smartly's **Pydantic AI**-based agentic orchestration harness, including progressive tool discovery, Plan Mode control flow and LLM-friendly tool interfaces, as well as Smartly's external MCP server.
- Maintaining **PostgreSQL** storage and **Redis** caching for agents' metadata, message history and draft outputs, plus PoCs for a persisted artifacts log; Redis also backs our **BullMQ** job queues.
- Unblocked 2 core teams through cross-team collaboration, shipping emergency features, bug fixes and refactors.
- Prototyped the company's 1st external MCP gateway and **TypeScript**-based MCP servers, creating general guidelines for good tool design and MCP implementation for 20+ engineering teams.
- Assisted in the 2-person migration of the company's Frontend monorepo of 4 million lines of TypeScript from **Jest** to **Vitest**, and solo-migrated 4 more repos from Jest/ESLint/Prettier to Vitest/Oxlint/Oxfmt.
- Optimized 2 existing LLM-based services through **LangFuse** trace monitoring and evals-based prompt optimization, reducing the cost of inference by 2.5x without loss of performance.
- Part of the company's AI Productivity Frontier group, delivering Knowledge Sharing sessions to 200+ R&D employees.

Musixmatch – ***Research Assistant*** — ***Feb 2026 – Aug 2026***

- Built an ETL data pipeline in Python to exhaustively collect, clean, and deduplicate over 4 million tracks' lyrics across languages and genres.
- Ran batch LLM annotation on 15,000+ Finnish tracks, cleaning up malformed LLM outputs into structured data.
- Replicated and extended Professor Eric Malmi's research on Finnish rhyme analysis and hip-hop networks with cross-linguistic and literary analysis. Blog

Aalto University – ***Master's Thesis Worker*** — ***Jan 2026 – Jul 2026***

- Developed feature-constrained gradient attacks and a confidence-triggered fallback defense for VR cybersickness classifiers using sensor-based LSTM models.

Nokia Solutions & Networks – ***R&D Trainee*** — ***Aug 2025 – Jan 2026***

- Synced 100,000+ documents across internal Nokia sources into **Vertex AI Search** on **GCP** with a **Python** ETL pipeline for RAG-based product queries.
- Built LLM-based internal tools in **Python** and **React** for logs processing and analysis, on a Gemini Agent Platform multi-agent harness combining fanout research, network logs triaging, and RAG-based QnA, reducing the incident triage time by ∼40%.

Aalto University – ***Teaching Assistant*** — ***Aug 2024 – 2025***

- Assisted in core Master's-level Security courses, totalling over 500 students, by holding office hours, designing network security assignments, and creating tools for fast grading.

Grinnell College (DASIL) – ***Research Assistant*** — ***Aug 2022 – May 2023***

- Built multivariate statistical models in **R** and **Python** from Stat2Labs game data (platform used by 1,000+ students across 5 institutions).

Microsec Zrt. – ***Software Engineer Trainee*** ***Jun – Aug 2022***

- Extended a PKI certificate linter in **Go** to handle X.509 extension edge cases following RFC 5280.

PROJECTS

Assorted Agentic Systems Projects ***2026 – Present***

- Participating in the Teleport Coding Challenge, using an agent swarm to port NetHack 5.0, a roguelike game with 500,000 lines of C and Lua code, to JavaScript with byte-exact behavior.
- Designing guardrails for multi-agent orchestration and a multi-turn evals framework to prevent common pitfalls with independent long-horizon agents like reward hacking, challenge bypass and goal misalignment.
- Building a personal assistant on Cloudflare Durable Objects with the Pi agent harness and OptChat-based memory: a context management system based on a summary tree of the entire chat history.

MLX Silicon Inference assorted projects ***2026***

- Ported Open Source Decision models into an MLX-compatible framework, including Cloudflare Clef. Evaluated and optimized inference performance on an M4 Pro with relevant Jev-based evals and benchmarks.
- In separate **llama.cpp** experiments using **Metal**, speculative decoding increased Gemma 4 12B throughput by 12.1% across three tasks.

Scalable Agentic Systems Evaluation ***Jan – Apr 2026***

- Benchmarked 300 software engineering tasks (SWE-Bench Lite) across 3 different harnesses, including mini-swe-agent and Claude Code, and 6 open-weight models like GLM-4.7.
- Deployed **vLLM** with **ROCm** on CSC's LUMI supercomputer (AMD MI250X GPUs) using **SLURM**; profiled evals results alongside GPU utilization, memory, latency, and concurrency, fixing incompatibilities with LUMI's **Singularity**-based containers, models' malformed reasoning, erroneous tool usage, and missing AMD support in inference cookbooks.